
SGAnalog: An End-to-End Circuit Benchmark from Open-Source Silicon Tapeouts

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Abstract

Existing analog integrated circuit design benchmarks make two questions hard to answer: whether a model has learned transferable circuit skills rather than recalled familiar examples, and whether its output works under defined process and test conditions. We introduce a benchmark built from human-designed, open-source circuits associated with Tiny Tapeout manufacturing shuttles. The collection contains 273 topologically distinct top-level designs.

Every source is retrieved at the revision recorded for its shuttle submission and processed in a fixed containerized environment. The pipeline exports each eligible schematic image and its SPICE netlist from the same source file, giving transcription an exact structural reference. Commit dates support model-specific training-cutoff analysis, while author testbenches provide the simulation context for sizing.

The benchmark evaluates schematic-to-netlist transcription and device sizing. Across seven models on a fixed set of 66 transcription tasks, the strongest model reaches 56.1% exact graph isomorphism, and six of seven models drop sharply from the small to the medium tier. For one frontier model, removing author-chosen labels reduces exact matches while preserving aggregate structural F1, suggesting that labels can aid connectivity tracing. On the 17 sizing tasks, the leading model converges on all 17 proposals and reaches 91.2 out of 100 against the human reference, while the two newest Claude models refuse 4 and 11 of the same prompts they transcribe without objection; a proposal without sizes scores zero. The two tasks produce different model rankings, exposing distinct visual and design capabilities and, in one family, a policy rather than capability limit.

1 Introduction

Analog integrated circuit design gives model outputs a concrete standard: the topology must be correct, and the sized circuit must work under defined process and test conditions. Evaluating these capabilities requires more than a collection of familiar circuit examples. The benchmark must also expose possible training overlap and provide an executable reference for function. Existing resources rarely provide both.

Separating capability from recall. Many analog benchmarks draw from canonical textbooks [1, 2], published circuit figures [3, 4], or synthetic enumeration [5]. These sources are useful, but their public availability makes prior model exposure difficult to rule out. Recent systems already approach saturation on the solved-task count of AnalogCoder [6, 7]. Without dated provenance and an exposure protocol, a high score does not reveal whether the model generalized or recalled a familiar design.

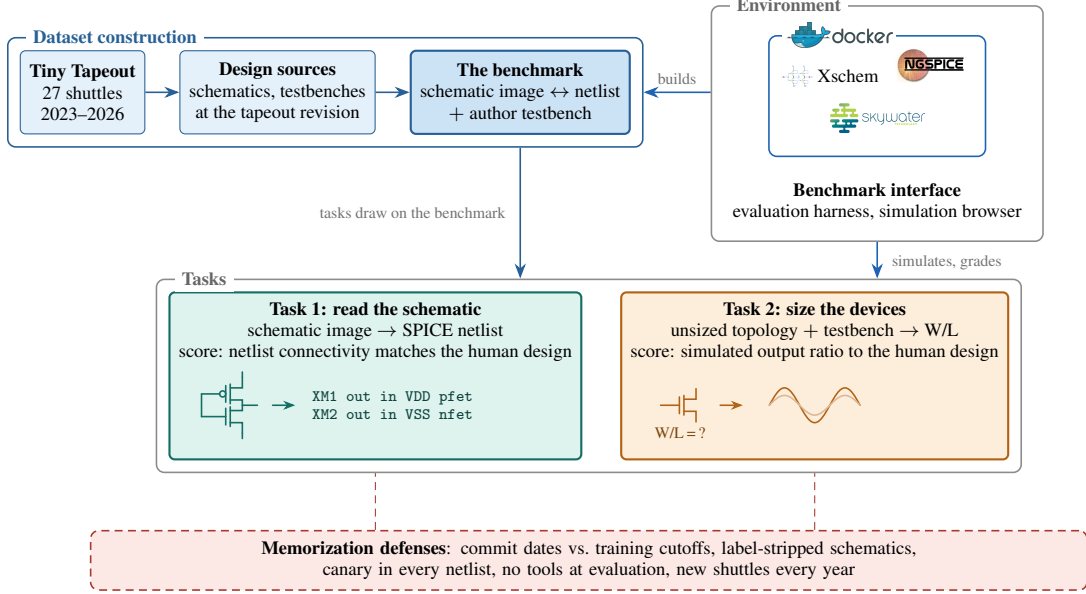


Figure 1: Benchmark overview. Versioned Tiny Tapeout sources are converted into matched schematic images and netlists, with author testbenches retained for simulation. Transcription is scored by device–net graph isomorphism, the comparison at the core of an LVS check; sizing is scored by simulation; and potential exposure is analyzed with commit-aware controls.

Verifying function. Large datasets often provide topology without device sizes, testbenches, or a process design kit (PDK). Simulator-backed benchmarks provide the missing context but cover a small set of base circuits and expand it through variants. The field therefore lacks a broad set of human designs that supports both structural comparison and execution under the original test conditions.

Why open-source tapeouts. Tiny Tapeout [8] provides a different foundation. Its machine-readable index links each shuttle submission to an exact repository revision. The analog repositories collectively contain transistor-level schematics, netlists, PDK context, and author testbenches. The recorded revisions make the pipeline reproducible, shuttle dates support model-specific cutoff analysis, and new shuttles provide a continuing refresh path. Verified open-source licenses allow the resulting artifacts to be redistributed. The collection spans amplifiers and comparators; ADC, TDC, and DAC blocks; oscillators and timing circuits; power, bias, and reference circuits; RF front ends and filters; and sensing interfaces. It also includes smaller passive, switching, and memory-related subcircuits. The fixed evaluation sets are eligibility-filtered subsets of this broader collection rather than equal-sized samples of every family.

Scope. The benchmark targets schematic-level reasoning, not layout or subjective design style. Transcription measures whether a model can recover circuit structure from an image. Sizing measures whether model-selected device dimensions preserve simulated behavior in the author’s testbench. Topology generation remains future work because it requires a uniform design brief and automatic testbench alignment.

Contributions. We call the benchmark SGAnalog, for silicon-grade analog.¹

1. A license-verified, commit-pinned benchmark from open-source Tiny Tapeout submissions, built with a deterministic containerized pipeline, graph-hash grouping, and aligned schematic images and netlists generated from the same source (§3).
2. Two end-to-end evaluations: graph-scored schematic transcription and simulator-scored device sizing against the human reference under the same PDK and author testbench (§4).

¹Code and data: <https://github.com/Li-Yueting/SGAnalog>.

3. An exposure-aware protocol combining commit dates with model-specific cutoffs, closed-book evaluation, blinded renders, release canaries, and refreshes from new shuttles (§5).
4. A seven-model baseline showing a broad small-to-medium transcription drop and different rankings across transcription and sizing (§4).

2 Related work

Table 1: Analog benchmark landscape. Sized = devices carry numeric sizes. TB = author testbenches are included. PDK = an open process design kit simulation flow is included. “part” marks partial coverage. Audit = the benchmark ships a contamination audit. Scale follows the unit reported by each source and is therefore not directly comparable.

Benchmark	Scale	Provenance	Sized	TB	PDK	Audit
AnalogGenie [4]	3,350 topo.	textbooks + papers	no	no	no	no
CktGNN / OCB [5]	10k op-amps	synthetic	part	part	no	no
AICircuit [9]	9 topo. / 361k pts	authored	yes	yes	part	no
OSIRIS [10]	5 templ. / 87.1k layouts	authored	yes	yes	yes	no
MAGICAL-CIRCUITS [11]	26 blocks	UT Austin silicon	yes	part	no	no
AnalogCoder [6]	24 tasks	authored	part	yes	no	no
AMSnet 1.0 / 2.0 [12]	734 / 2,686	textbook scans	no	no	no	no
Masala-CHAI [13]	7,500	textbook scans	no	no	no	no
AnalogGym [3]	30 topo.	published papers	part	part	part	no
SGAnalog (ours)	273 distinct topologies	Tiny Tapeout repos	yes	part	yes	yes

Prior image-based sets such as AMSnet [12] and Masala-CHAI [13] pair scanned textbook figures with netlists produced by manual or model-assisted extraction. In this benchmark, every eligible image–netlist pair is generated from one schematic source, avoiding a separate annotation step.

AnalogGym [3] is the closest simulator-backed benchmark in intent. Its amplifier topologies come from published literature, whereas our sizing reference is the submission author’s design evaluated under the same PDK and testbench as the model. Concurrent work, NotSoTiny [14], uses the digital side of Tiny Tapeout [8] for RTL code generation; our benchmark targets the analog artifacts and their simulation context.

3 Dataset construction

Harvest funnel. Table 2 walks the index down to the fixed task sets; Appendix B holds the full triage, licensing, and composition inventory. Blobless clones triage every candidate by file tree and license without downloading design data, licensing is verified from the license texts as fetched at the pinned commits rather than assumed, and design source is fetched at the commit the shuttle index records, never at HEAD.

Circuit identification. Counting circuits naively fails in both directions: every .sch overcounts (symbols, testbenches, sub-blocks); counting only hierarchy tops undercounts, because the standard xschem [15] idiom places the interesting circuit under its testbench. We classify by role instead. A *testbench* contains stimulus and simulation control, or matches the naming conventions for one. A *circuit* is any non-testbench schematic. A *top-level design* is a circuit that no other non-testbench schematic instantiates. Table 4 (Appendix B) gives the resulting composition by role, tier, and era. The trivial tier also includes transistor-free passive networks and device models such as photodiode equivalents.

Redundancy. Repository counts overstate the diversity of the tapeout-derived collection because blocks are resubmitted and forks preserve earlier circuits. We compute a strict Weisfeiler–Lehman hash of each top-level device–net bipartite graph after removing net names. Repeated pad models, inverters, tutorial circuits, and R-2R blocks account for most shared hashes. The resulting candidate duplicate groups are released for split control. Task 1 scoring uses explicit graph isomorphism rather than hash equality.

Table 2: Dataset spine: from the shuttle index to the fixed task sets. Appendix B gives the full triage, licensing, composition, and redundancy inventory.

From index to task sets	<i>n</i>
Index projects (27 shuttles)	4,572
Entries with analog pins	237
Repositories contributing circuits	125
Schematics fetched at pinned commits	1,448
Circuits	933
Top-level designs	425
Unique device–net graphs	273
Fixed tasks: transcription / sizing	66 / 17

Reproducibility contract. Duplicate file stems and archived copies can make hierarchy resolution depend on filesystem order, so candidates are resolved by depth and then full path. Every stage, from harvest through simulation, executes inside one pinned container image (IIC-OSIC-TOOLS [17], digest `sha256:ae380e2b...df38`), which pins ngspice [16] and supplies the open PDKs; the host requires only Docker. Classification tables from two complete runs are byte-identical, and simulation tables regenerate identically up to their per-run timing columns.

Task-set construction. A curated core spans circuit classes, difficulty tiers, and eras. For transcription, leaf schematics with every symbol resolved are eligible; deterministic round-robin sampling over era and tier fixes 66 tasks. For sizing, we audit 412 author-testbench runs after deterministic include-path repair, without changing circuits or stimuli. A run is eligible only if it completes within 600 seconds, contains real device cards, and resolves the schematic hierarchy; 190 runs meet this gate. End-to-end device-under-test mapping and the primitive-only hierarchy requirement then yield 17 tasks. Both sets remain fixed across models. The release ships a browser interface over the verified circuit–testbench pairs (Appendix C). Hierarchical sizing tasks await the measurement layer described in Section 6.

4 Tasks and evaluation

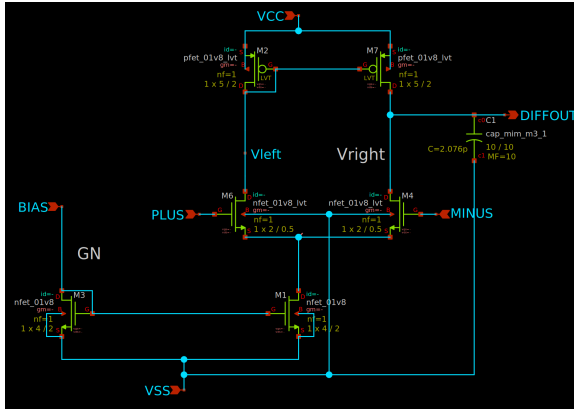
The benchmark defines two complementary tasks. Transcription tests schematic reading; sizing tests device choices under simulation.

Task 1: schematic-to-netlist transcription. Input: the rendered schematic (Figure 2). Output: a netlist. Scoring is by graph isomorphism, never by text match. We build a device-net bipartite multi-graph from each netlist and test it for isomorphism against the ground truth, treating the drain and source of a MOSFET as interchangeable and ignoring net names. We report exact isomorphism, a structural F1 over device and net signatures for partial credit, and W/L parameter accuracy. Because image and netlist derive from the same `.sch` file, each image corresponds exactly to its reference netlist, and render style becomes a controllable nuisance variable.

Transcription is an established reading task rather than a design task; AMSnet [12] and Masala-CHAI [13] study the same extraction problem. Here it also provides a common evaluation across circuits regardless of testbench state, enabling the commit-date analysis below.

The baseline below uses the default render. The release also ships a blinded render for each schematic. Device terminals, size annotations, and PDK model names remain because reading them is part of the task. Universal supply names also remain because they carry little circuit-specific retrieval value. Author-chosen net labels and instance names are removed as plausible retrieval handles; removal is driven by each circuit’s netlist rather than by pattern matching. The blinded task stays well posed because the scorer ignores names.

On `claude-opus-5`, blinding lowers exact isomorphism from 51.5% to 36.4% while structural F1 is unchanged and parameter accuracy improves. Because the scorer ignores names, the result is consistent with labels serving as visual anchors for connectivity rather than as part of the answer itself. It does not rule out prior exposure; the cutoff analysis addresses that separate channel.



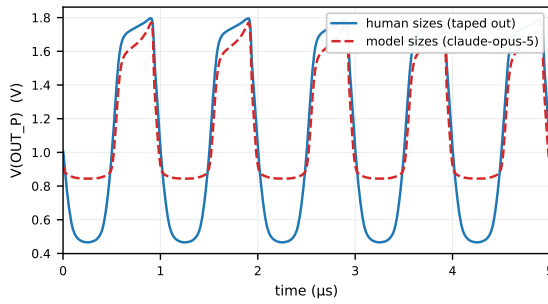
* Differential amplifier
* (5T OTA with output cap)

```

XM2 Vleft Vleft VCC VCC
+ sky130_fd_pr__pfet_01v8_lvt
+ L=2 W=5 nf=1
XM7 DIFFOUT Vleft VCC VCC
+ sky130_fd_pr__pfet_01v8_lvt
+ L=2 W=5 nf=1
XM6 Vleft PLUS TAIL VSS
+ sky130_fd_pr__nfet_01v8_lvt
+ L=0.5 W=2 nf=1
XM4 DIFFOUT MINUS TAIL VSS
+ sky130_fd_pr__nfet_01v8_lvt
+ L=0.5 W=2 nf=1
XM3 BIAS BIAS VSS VSS
+ sky130_fd_pr__nfet_01v8 L=2 W=4 nf=1
XM1 TAIL BIAS VSS VSS
+ sky130_fd_pr__nfet_01v8 L=2 W=4 nf=1
XC1 DIFFOUT VSS
+ sky130_fd_pr__cap_mim_m3_1
+ W=10 L=10 MF=10 C=2.076p

```

Scored: graph-isomorphic; W/L accuracy 1.00.



Device	Human	Model
input pair	3.6/0.38 × 2	20/0.5 × 10
loads (PMOS)	3.6/0.38 × 2	5/0.5 × 2
tail	3.6/0.38 × 2	7.2/0.38 × 4
bias mirror	3.6/0.38 × 2	3.6/0.38 × 2
bias resistor	L=1.85 × 10	L=30 × 1

Sizes as W/L in $\mu\text{m} \times \text{nf}$. In the same testbench the human sizing swings ± 665 mV at the output; the model's reaches ± 463 mV with a compressed lower rail. The delta is what simulation-based scoring measures.

Figure 2: Both tasks, end to end. Top: transcription. The input schematic image (left) and claude-opus-5’s verbatim output netlist (right) for a five-transistor OTA. Its net names differ from the drawing, but its graph is isomorphic to the reference. Bottom: sizing for a fully differential amplifier. The model receives the topology with sizes removed, then its proposal and the human reference run in the same author testbench. Aggregate results appear in Table 3.

Task 2: device sizing. Input: the human topology with W/L stripped to placeholders, the testbench, the PDK, and target metrics. Output: sizes; the topology is fixed. The reference is the simulated performance of the human-authored sizing under the same PDK, testbench, and simulator. Figure 2 (bottom) shows the task executed end to end on one circuit: the model’s proposed sizes beside the reference sizes, and both simulated in the author’s unmodified testbench.

The task runs on the fixed sizing set (§3); scoring is author-testbench-relative. Both sizings run the author’s analyses. The reference selects the output vector with the largest transient swing or absolute low-frequency AC gain, and the model is evaluated on that same vector. For each eligible metric, let r be the model-to-reference ratio; the metric score is $\min(r, 1/r)$, averaged across available metrics. A nonconvergent run scores zero, while a convergent task without an eligible scalar metric receives a survival score of one. A proposal that provides no sizes also scores zero: the unmodified deck holds the human sizes, so simulating it would grade the reference against itself. Table 3 reports the mean scores, and Figure 3b shows every cell behind them. Every Claude or Gemini proposal that provides sizes converges, whereas both GPT generations produce several invalid device geometries. Convergence alone is too weak a bar: on one task, two proposals converge with output swing below one percent of the reference, a failure the convergence count cannot see and the simulated metric does. Refusal is a third failure mode, and it lands on the newest models: claude-opus-5 returns an empty API-level refusal on 4 of the 17 prompts and claude-fable-5 on 11, while both transcribe the same circuits without objection. A full repeat run of claude-fable-5 refuses 12, ten prompts refused both times and three changing sides, so the refusal set is stable in bulk but not per prompt. On the tasks they answer, the two score 97.2 and 90.2. The sizing ranking therefore diverges from transcription’s (Figure 3c) for two reasons: sizing knowledge is text-borne and present at every Claude tier, and on the newest models a safety policy, not capability, sets the score. Standardized

per-class measurement templates are the next refinement (§6). Because the human sizes are public, Section 5 treats answer-key exposure explicitly.

Evaluation protocol. The released harness evaluates any model behind an API in a single invocation, and every request declares no tools, so evaluations are closed book by construction. Outputs produced outside the harness are scored from a standard submission format against the frozen task list, and our own published runs pass through that same path, so submitted and published results are graded by identical code. The scorer is deterministic: identical inputs produce identical result tables across processes and hash seeds, a property the parameter metric was corrected to guarantee.

Baseline. Table 3 reports aggregate results, and Figure 3 shows the per-task evidence behind them.

Six of seven models drop sharply between the small and medium tiers (Figure 4, Appendix A); Haiku is already at floor on the small tier. The failure among the remaining models is wiring, not reading: a model that recovers a topology also reads its W/L annotations nearly perfectly, so a single blended score would hide which skill is missing. Tier difficulty is also not strictly monotonic: Opus and Gemini score higher on small circuits than on trivial ones. The trivial tier is dominated by passive networks and device models, whose vocabulary produces errors in models that transcribe a five-transistor amplifier cleanly.

Failure mode separates the leading models as much as exact accuracy does (Figure 3a). Claude-fable-5 leads both exact matches and structural F1. Behind it the frontier splits: claude-opus-5 is more often fully correct than gpt-5.6-sol, but its failures tend to be severe, while sol produces repairable near misses and holds the second-best structural F1 despite a lower exact-match rate. The preferable model therefore depends on whether a downstream system can repair a netlist.

The scorer itself required two corrections, reported because each would have distorted a leaderboard. Parameter accuracy once scored against one arbitrary isomorphism of each symmetric circuit; it now pairs devices under the mapping with maximum parameter agreement among structurally valid isomorphisms, selected deterministically rather than by hash-seed order. PDK resistor terminals were once ordered pins, although the symbol draws no orientation and pin order cannot be read from the image. The corrected scorer still rejects genuinely misread connections, such as visually explicit bulk ties replaced by rail defaults. Table 3 also reports model-specific cutoff splits; Section 5 states what can and cannot be inferred from them.

Table 3: Model results on the fixed transcription set. Cutoffs are vendor-reported; post n counts tasks whose pinned revision postdates the cutoff, ruling out that exact revision from the model’s pretraining window. Post/pre columns give exact isomorphism on each side. Sizing is the Task 2 score on a 100-point scale, with a proposal that provides no sizes scoring zero; No ans. counts those proposals (refusals, one unparseable answer) out of 17.

Model	Cutoff	Iso	Struct	W/L	Post n	Post / pre iso	Sizing	No ans.
claude-fable-5	2026-01	56.1%	0.853	0.921	13	30.8% / 62.3%	31.8	11
claude-opus-5	2026-05	51.5%	0.711	0.929	0	—	74.3	4
gpt-5.6-sol	2026-02	42.4%	0.831	0.942	13	30.8% / 45.3%	78.2	0
gemini-3.1-pro-preview	2025-01	39.4%	0.767	0.977	36	38.9% / 40.0%	91.2	0
claude-sonnet-5	2026-01	19.7%	0.604	1.000	13	7.7% / 22.6%	85.4	0
gpt-5	2024-09	13.6%	0.512	0.988	38	15.8% / 10.7%	79.0	0
claude-haiku-4-5	2025-02	1.5%	0.307	1.000	35	floor	78.3	1

5 Measuring contamination

Public GitHub repositories may enter model pretraining, so this benchmark cannot assume independence. It instead provides circuit-level evidence for probing potential exposure: pinned commit dates, blinded renders, release canaries, and a closed-book protocol.

Commit-date analysis. Each circuit retains its shuttle and pinned commit date, which can be compared with a model’s reported training cutoff. Table 3 performs this comparison per model. Claude-opus-5 has no post-cutoff tasks in the fixed set. Across the two usable older-cutoff comparisons, exact-match accuracy shows no consistent pre-cutoff advantage. This descriptive comparison

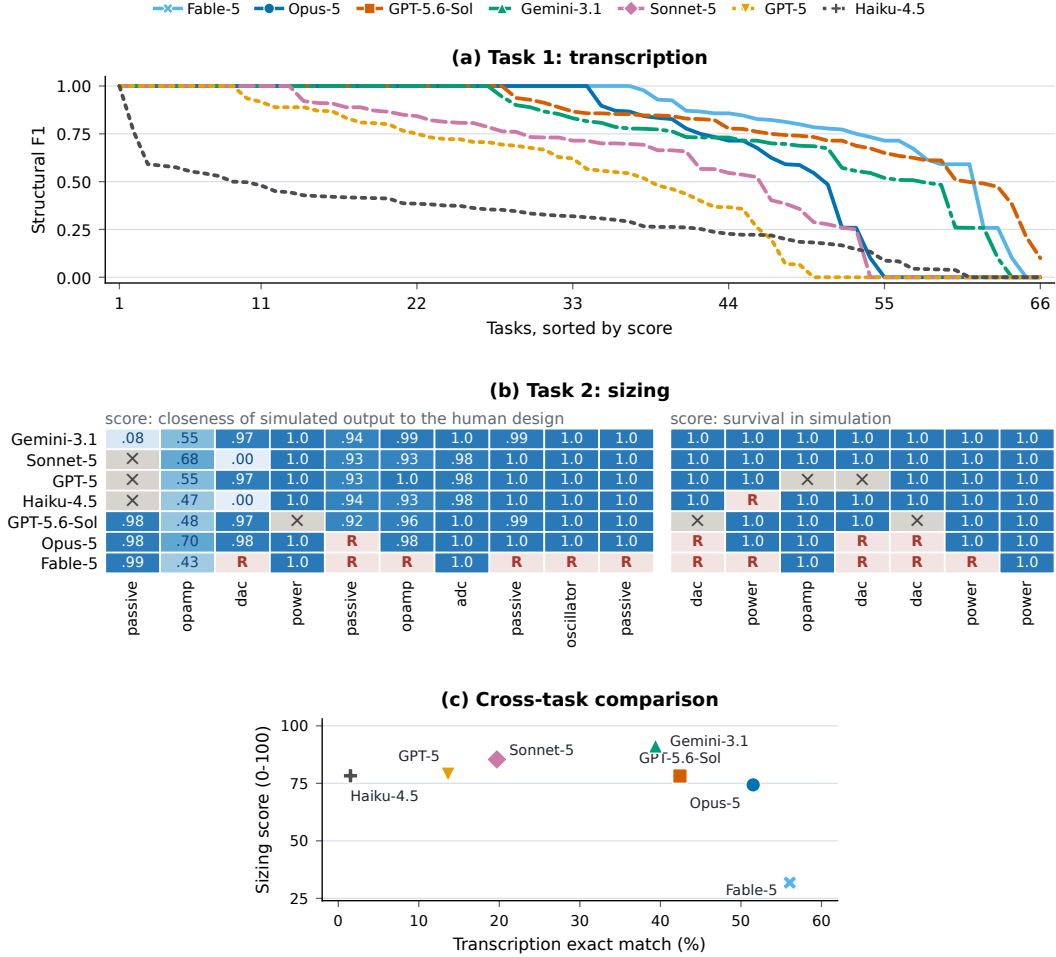


Figure 3: Main results for both tasks. (a) Task 1: each model’s 66 structural F1 scores sorted in descending order. The width of the plateau at 1.0 equals the exact-match count, and the tail shape is the failure mode: Opus falls steeply once it fails, while Sol declines gradually. The encoding uses no banding thresholds. (b) Task 2: the complete score matrix, seven models by 17 sizing tasks, one cell per proposal. Each cell prints its score, with blue depth as a redundant encoding; a gray cross marks a nonconvergent sizing, and a red R marks a proposal that provided no sizes, which scores zero. The left group is scored by how close the proposal’s simulated output comes to the human design’s, hardest answered task first; in the right group the author analyses expose no usable scalar, so a convergent sizing scores one. The two .00 cells, on a DAC task, are convergent sizings whose output collapsed. (c) Transcription accuracy versus sizing score: model rankings differ across the two capabilities. The sizing axis is truncated to keep the seven points legible. The difficulty-tier breakdown of Task 1 is in Appendix A.

is not a contamination estimate: difficulty differs across the date boundary, graph-hash groups are not separated across it, and an earlier version of a design may predate the pinned revision. The commit metadata supports the same analysis for future models, while new shuttles expand the post-cutoff side.

Training-time and inference-time channels. Commit dates probe possible exposure in training data. A model evaluated with tool access has a second channel: the upstream repositories are public, so a search-enabled agent can retrieve an answer at inference time regardless of era. The protocol separates the two. Baseline scores are closed-book: evaluation requests declare no tools, so the model has no retrieval channel by construction. Systems with search or repository access belong to a separate open-book track and must be labeled as such. The blinded render removes author-

selected labels from the image. Reproducing an invisible internal name is evidence of memorization or external retrieval, so the blinded condition also provides an exposure diagnostic for submitted results.

Answer-key exposure differs by task. For transcription the artifact given to the model is an image that does not contain the netlist. For sizing the upstream repository contains the human sizes, which is why sizing is scored at perturbed operating points with regurgitation diagnostics.

Canary. Every released netlist carries the BIG-bench canary as an ignored SPICE comment, and the release check verifies its placement. The canary marks generated artifacts; it cannot protect upstream sources that were already public. Commit-date splits address that channel, while blinded renders address inference-time retrieval. Per-circuit memorization probes remain an archival milestone.

6 Limitations and roadmap

Most entries target Sky130 [18]; broader coverage of GF180 and IHP SG13G2, including SiGe, is planned from MPW-mirror, IHP, and ISHI-KAI sources. Schematics with unresolved host paths are excluded from rendering-sensitive splits. Circuit-class labels are keyword-derived and will be reclassified from netlist graphs. Top-level designs without a parseable netlist remain outside the graph-hash analysis.

Three milestones remain for the archival version. First, because author testbenches expose heterogeneous measurements, we will add per-class templates over the eligible set, starting with gain, gain-bandwidth product, and phase margin for the amplifier class. This layer widens sizing coverage beyond the leaf-subcircuit set. Second, the per-circuit memorization probes described in Section 5, joining the delivered blinded-render comparison. Third, topology generation as a task. We do not claim it in this version because two of its components are unsolved: a uniform representation for the design brief, and automatic pin-matching between a generated circuit and the author testbench. Both build on the measurement layer; claiming the task before its scoring exists would overstate readiness.

Acknowledgments

We thank the Tiny Tapeout community and the designers whose open-source circuits form this benchmark, and the AI4ChipDesign reviewers for their feedback.

Ethics, licensing, and attribution

All redistributed designs are Apache-2.0, verified from license texts archived at the pinned commits; the released attribution file credits every repository and its authors, and per-circuit manifests carry source URL, commit, license, and extraction method. The Tiny Tapeout index [8] is CC0; the PDKs are Apache-2.0. Benchmark code and released data use Apache-2.0, matching the source licenses. The code and data are released at <https://github.com/Li-Yueting/SGAnalog>, together with a simulation browser that replays precomputed runs of the verified pairs. Designer names appear only as attribution of their public, licensed work. Per the workshop’s LLM policy: an LLM assistant was used in building the harvest and evaluation harness and in drafting this manuscript; the authors verified every released number against the committed tables, and the release ships the consistency checker that enforces the match.

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A Difficulty-tier breakdown of Task 1

Figure 4 plots exact isomorphism per difficulty tier for the seven evaluated models, the per-tier view behind the tier claims in Section 4.

B Full dataset inventory

Table 4 expands the spine of Table 2 into the complete harvest, licensing, composition, and redundancy record.

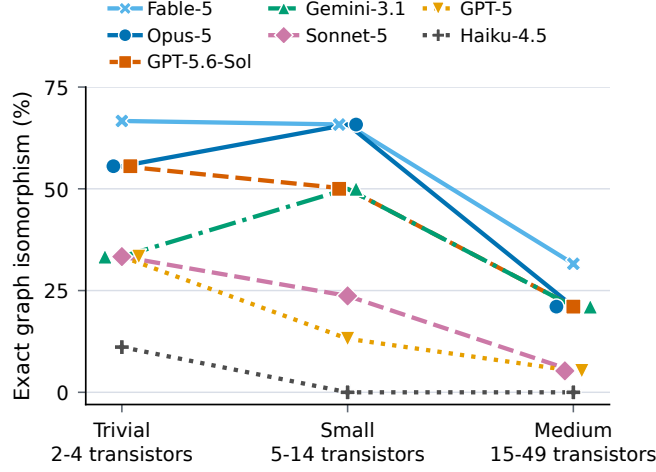


Figure 4: Exact isomorphism by difficulty tier ($n = 9, 38, 19$): five models drop from small to medium, while Haiku is already at floor; Opus and Gemini improve from trivial to small. Markers with identical scores are offset slightly within each tier for visibility; only their vertical position encodes accuracy.

Table 4: The full dataset inventory. Left: the harvest funnel with triage and licensing detail. Right: composition by role, transistor-count tier, era, and device-net graph hash.

Harvest funnel	n	Composition	n
Index projects (27 shuttles)	4,572	Circuits	933
Entries with analog pins	237	with a PDK transistor	809
Unique repositories	195	top-level designs	425
ship xschem [15] schematics	133	sub-blocks	508
ship SPICE netlists only	12	Testbenches	515
layout data only (unusable)	48	circuits driven by one	350
fail to clone	2	Top-level tiers: trivial (0–4 transistors)	164
Usable repositories	145	small (5–14)	161
license verified Apache-2.0	144	medium (15–49)	78
hold fetchable schematics	129	large (50–149)	20
contribute classified circuits	125	very large (150+)	2
Schematics fetched at pinned commits	1,448	non-trivial (5 or more)	261
		Eras: 2023–24 / 2025 / 2026	194 / 143 / 88
		Graph-hashable top-level designs	357
		unique device-net graph hashes	273
		redundant designs	84
		duplicate groups / cross-repository	43 / 32
		Fixed tasks: transcription / sizing	66 / 17

C Simulation browser

The release includes a local browser interface over the verified circuit–testbench pairs (Figure 5). For each pair it shows the circuit and testbench schematics, both extracted ngspice netlists, the upstream repository at its pinned commit, and any specifications the project’s own documentation publishes. A run started from the page executes through the same pinned-container entry point as the pipeline, so its result is bit-identical to the batch simulatability check, and the waveforms are parsed from the run’s raw files. The sizing task set of Section 3 is drawn from these verified pairs. The released repository builds a static companion page with precomputed runs of ten representative pairs, one per circuit class; the live browser ships with the repository and requires only Docker.

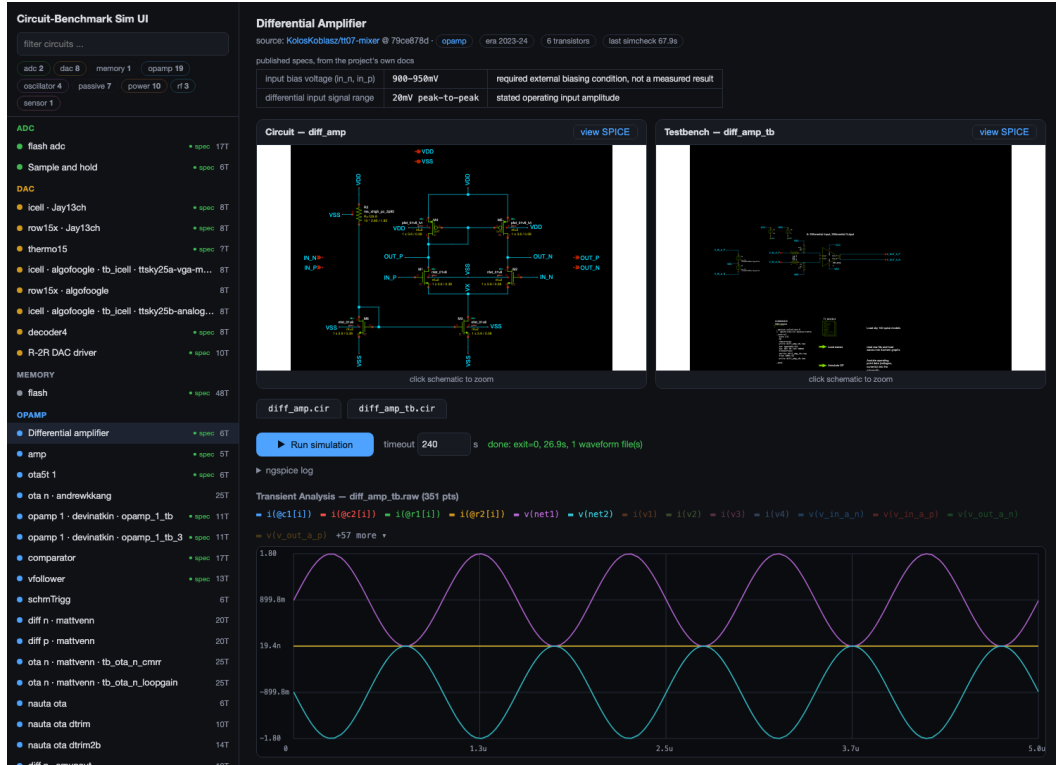


Figure 5: The simulation browser on one verified pair, a differential amplifier from the corpus: published specifications, circuit and testbench schematics, and the transient waveforms produced by a run inside the pinned container (exit 0 in 26.9 seconds; the two complementary output sines around the input common mode).